THE MASTERMIND COACHING CLASSES

CBSE / RBSE • NURSERY – CLASS XII • JAIPUR

Road No. 5, VKIA, Jaipur — 9887258679

ANSWER KEY & MARKING SCHEME SET 07

Full Syllabus (Chapters 1–13) | Class 10 Science | Maximum Marks: 80

SECTION — A (Objective Type) — Answers — 20 × 1 = 20

Q.	Correct Option	Brief Reasoning
1	**(a)** Fe_2O_3 and aluminium	$Fe_2O_3 + 2\,Al \longrightarrow Al_2O_3 + 2\,Fe$
2	**(d)** CO_2	Lime water turns milky
3	**(a)** evaporation	Salt pans
4	**(d)** gangue	Removed during enrichment
5	**(b)** Student A is correct	ZnO reacts with both
6	**(b)** long-chain carboxylic acids	Fatty acids
7	**(b)** naphthalene	High carbon content
8	**(a)** blood capillaries	In Bowman's capsule
9	**(b)** ovaries	Female hormone
10	**(b)** 46	Diploid
11	**(c)** zygote	Diploid cell
12	**(a)** Correct	Haploid
13	**(d)** seven	Height, colour, shape…
14	**(b)** 3×10^8 m/s	c
15	**(a)** refraction, dispersion and internal reflection	Raindrop as prism
16	**(b)** 2 kWh	0.5×4
17	**(a)** 1 C/s	$I = Q/t$
18	**(a)** heating effect of current	Melts
19	**(d)** heat	Respiration
20	**(b)** abiotic components	Non-living

SECTION — B (Very Short Answer) — Answers — 8 × 2 = 16

Q.	Expected Answer / Value Points	Marks
21	It decomposes in light: $2\,H_2O_2 \longrightarrow 2\,H_2O + O_2$; dark bottles prevent this.	1+1
22	(i) tartaric acid (ii) citric acid	1+1
23	Water has much less dissolved oxygen than air, so aquatic animals breathe faster.	2
24	False — hunger is controlled by the hypothalamus; the cerebellum controls balance.	2
25	They store food for the developing embryo.	2
26	The two alleles of a gene separate during gamete formation, so each gamete gets one.	2
27	It always gives an erect, diminished image and has a wider field of view.	2
28	It switches off the circuit automatically when the current exceeds a safe limit.	2

SECTION — C (Short Answer) — Answers — 8 × 3 = 24

Q.	Expected Answer / Value Points	Marks
29	Ions of the two reactants are exchanged: $NaOH + HCl \longrightarrow NaCl + H_2O$; Na^+ pairs with Cl^- and H^+ with OH^-.	1+1+1
30	Connect a metal wire/strip in a circuit with a cell and bulb; the bulb glows, showing current passes through the metal.	3
31	High blood sugar → pancreas secretes insulin → sugar falls → insulin release reduces. This prevents excess or deficiency.	3
32	(i) carries egg; site of fertilisation (ii) implantation and development of embryo (iii) nutrition, gas exchange and waste removal for embryo	1+1+1
33	$u = -25$ cm, $v = -100$ cm: $\frac{1}{f} = \frac{1}{v} - \frac{1}{u} = -\frac{1}{100} + \frac{1}{25} = \frac{3}{100} \Rightarrow f = 33.3$ cm; $P = +3$ D	3
34	Safety device of a low-melting wire in series with the live wire; excessive current heats and melts it, breaking the circuit	1+2

35	(i) acid turns it red; base no change (ii) base turns it blue; acid no change (iii) acid — red, base — yellow	1+1+1
	OR: Acid + base → salt + water, e.g. $\mathrm{NaOH + HCl \longrightarrow NaCl + H_2O}$; antacid relieving indigestion / baking soda on an ant sting	1+1+1
36	Removing producers kills all other levels; removing top carnivores increases herbivores which overgraze; impact depends on position	3
	OR: Accumulation of non-biodegradable chemicals at increasing levels in each trophic level; humans occupy the top of food chains	2+1

SECTION — D (Long Answer) — Answers

4 × 5 = 20

Q.	Expected Answer / Value Points	Marks
37	(i) distances from optical centre; along incident light +; heights above axis + (ii) $\frac{1}{u} = \frac{1}{v} - \frac{1}{f} = -\frac{1}{15} + \frac{1}{20} = -\frac{1}{60} \Rightarrow u = -60$ cm	2+3
	OR: (i) object on left, distances from pole; along incident light +, opposite −; heights above axis +, below − (ii) $f = R/2 = +16$ cm (iii) wide field of view; erect image	3+1+1
38	(i) A: ethanoic acid, B: ethanol, C: ethyl ethanoate (ii) $\mathrm{2\,CH_3COOH + Na_2CO_3 \longrightarrow 2\,CH_3COONa + H_2O + CO_2}$ (iii) esterification (iv) saponification: ethanol + sodium ethanoate	2+1+1+1
	OR: (i) $\mathrm{CH_3CH_2OH \xrightarrow{alk\cdot KMnO_4+heat} CH_3COOH}$ (ii) substances that add oxygen: alkaline $\mathrm{KMnO_4}$, acidified $\mathrm{K_2Cr_2O_7}$ (iii) combustion is burning in oxygen with heat and light; oxidation here converts alcohol to acid without burning	2+2+1
39	(i) 2000 + 300 + 100 = 2400 W (ii) $I = 2400/220 = 10.9$ A (iii) 15 A (iv) $2.4 \times 3 = 7.2$ kWh (v) each gets 220 V; independent working	1+1+1+1+1
	OR: (i) series (ii) parallel (iii) series: $I = 10/20 = 0.5$ A; parallel: total $R = 5\ \Omega$, $I = 2$ A (1 A in each) (iv) series: other goes off; parallel: other keeps glowing	1+1+2+1
40	(i) body → vena cava → RA → RV → pulmonary artery → lungs → pulmonary vein → LA → LV → aorta → body (ii) systole: contraction; diastole: relaxation	3+2
	OR: (i) receives bile (emulsifies fats) and pancreatic juice; intestinal juice completes digestion: proteins → amino acids, carbohydrates → glucose, fats → fatty acids + glycerol (ii) finger-like projections that increase surface area for absorption (iii) trypsin, lipase (amylase)	2+2+1

Evaluator's note: Award step marks as shown; accept any correct alternative method. In internal-choice questions, evaluate only the option attempted. Deduct up to ½ mark for missing units or figures where required.

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